

Report

Part 1

We were asked to develop a simplified version of the linear regression in C++. We used our existing java classes to help us implement the program in C++. The modification were: we left only `Dataset` class and deleted the standardized one and all related methods. Therefore, we didn't have to train any supervised learner model on standardized dataset and had to check only 2 models in `TestLearner` class.

Part 2

There were no possible alternative solutions discussed, as we were basically translating java code to C++ code. We used abstract class `Algorithm` to implement 2 gradient descent algorithms. Also, we used method overloading and overriding.

Part 3

The solution performed better than in java implementation as the computed value was exactly the same as the expected one. The parameter vector obtained corresponded exactly to the actual one, except for the bias term, which was very close to 0 (10^{-6} and 10^{-15}).

We didn't find any difficulties in the implementation of this Lab.